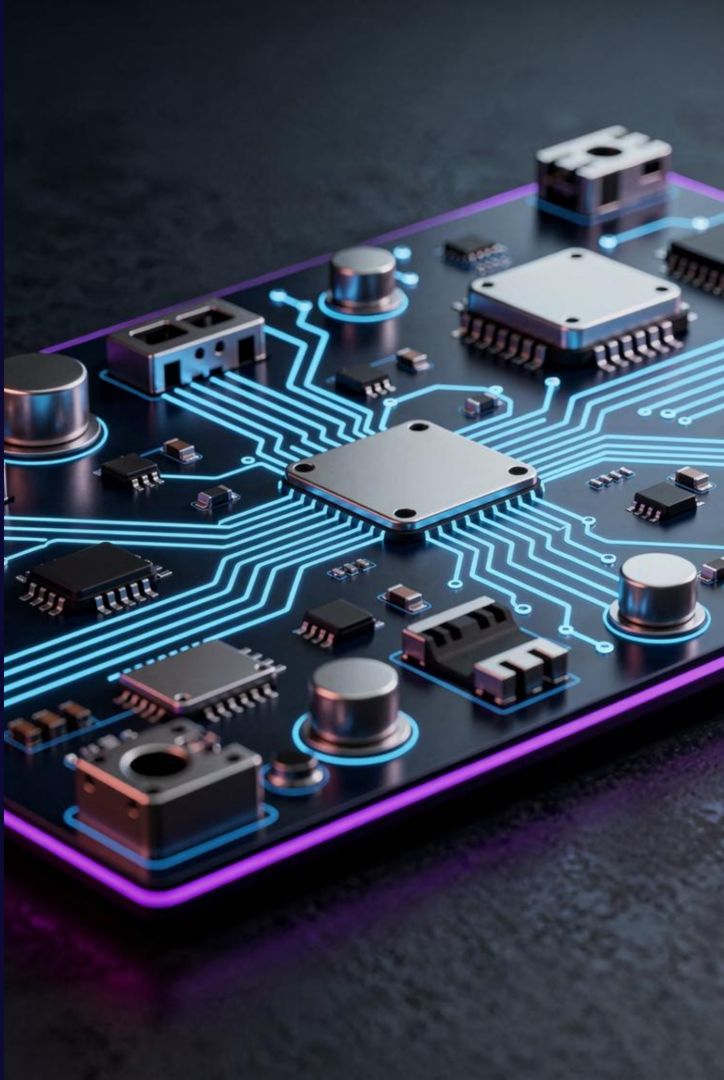
Caleido Xenia Device Engineering Playbook

ESG + AI + Connected Facility Systems

BOARD-GRADE ENGINEERING COMMAND CENTER



System Architecture Overview

The Caleido Xenia platform is built on a disciplined four-layer architecture — each domain isolated for signal integrity, compliance, and AI-readiness. This hierarchy ensures clean separation of concerns across power, sensing, control, and intelligence.

Layer 1 — Power & Energy Domain

Regulated power distribution, supercapacitor backup, energy telemetry, and fail-safe power sequencing for all downstream subsystems.

Layer 2 — ESG Sensing Layer

Air quality (PM2.5, CO_2, VOC), occupancy detection, and ambient environmental monitoring — the credibility foundation of the ESG platform.

Layer 3 — Control & Actuation

Smart lock motors, HVAC relay control, and protected actuation circuits with flyback suppression and LC filtering.

Layer 4 — Edge Intelligence

Cielo Epic AI integration, on-device ML inference, and HTNG/FIAS protocol readiness for hospitality-grade facility management.

PCB Partitioning Strategy

Physical partitioning is the first line of defense against signal degradation and EMI coupling. Each domain occupies a defined zone — noise crossover between zones is strictly prohibited by design rule.

Power Zone

High-current traces, regulators, and supercap bank. Isolated from analog with copper pours and keep-out regions.

Analog Zone

Sensor front-ends, ADC inputs, and reference voltage rails. Maximum isolation from switching noise sources.

Digital Zone

MCU, memory, edge AI processor, and communication modules. Contained clock domains with stitching vias.

Actuation Zone

Lock motors, relay drivers, and flyback protection. Fully isolated return paths to prevent ground contamination.

Design Rule: No noise crossover between zones. Violating this rule invalidates ESG sensor accuracy and EMI compliance simultaneously.

ESG Sensor Integrity — The Platform Differentiator

CRITICAL DIFFERENTIATOR

Hardware accuracy equals ESG credibility. The precision of PM2.5, CO_2, and VOC readings directly determines the defensibility of sustainability reporting and regulatory submissions. These are not commodity sensors — they are compliance instruments.

Sensor Placement Rules

- Minimum 15mm clearance from heat-generating components
- Dedicated airflow apertures aligned to sensor intake ports
- No obstruction within 20mm of PM2.5 optical cavity
- CO_2 sensors positioned away from board exhaust zones
- VOC sensors isolated from flux residue and solder mask outgassing areas

Thermal & Airflow Considerations

- Thermal vias under regulators must not channel heat toward sensing zone
- Airflow modeling required pre-layout sign-off
- Operating temperature validated across full -10°C to +60°C range
- Humidity compensation coefficients applied in firmware calibration layer

Power Architecture & Supercapacitor Strategy

The Caleido Xenia power subsystem is engineered for resilience — ensuring fail-safe lock operation during grid interruption, continuous ESG telemetry, and hybrid energy sourcing with full observability for sustainability dashboards.

Supercapacitor Fail-Safe

Supercap bank provides minimum 30-second hold-up for lock release on power loss. Sized to complete a full lock cycle with 20% safety margin. No battery disposal liability.

ESG Energy Telemetry

Per-circuit current sensing via precision shunts feeds real-time consumption data to the ESG dashboard. Enables sub-device energy attribution for sustainability reporting.

Battery + Grid Hybrid

Seamless transition between mains, PoE, and Li-Ion backup. Priority sequencing logic ensures critical loads — locks and sensors — are last to lose power.

Overcurrent & Reverse Protection

TVS diodes, polyfuses, and reverse polarity protection on all external interfaces. Designed to meet IEC 62368-1 and hospitality-grade service requirements.

Noise Management & Grounding Architecture

Signal integrity at the ESG sensing layer is non-negotiable. A disciplined grounding strategy eliminates common-mode interference that would corrupt sub-ppm VOC and particulate readings — and undermine regulatory credibility.

Solid Uninterrupted Ground Plane

A continuous copper ground plane on Layer 2 provides low-impedance return paths for all domains. No slots, cutouts, or routing violations permitted. Plane integrity verified via Gerber DRC before fabrication sign-off.

Analog Domain Isolation

Analog ground sub-domain connected to digital ground at a single star point beneath the ADC. Ferrite bead isolation on VCC rails feeding sensor front-ends. Guard rings applied around high-impedance nodes.

No Return Path Breaks

Every high-speed or high-current signal trace must have an uninterrupted return path directly beneath it on the ground plane. Any routing that crosses a plane split triggers an automatic DRC failure — enforced in Altium constraint manager.

Connectivity & Edge AI Integration

The Caleido Xenia connectivity stack is engineered to serve both the real-time control plane and the long-range telemetry plane — with full protocol readiness for hospitality industry standards and on-device AI inference via Cielo Epic.

Wireless Stack

WiFi 6 (802.11ax)

Primary cloud uplink for ESG data, OTA firmware, and remote management.

BLE 5.2

Mobile credential proximity, guest room keying, and commissioning interface.

Zigbee 3.0

Low-power mesh for distributed sensor nodes and HVAC actuator coordination.

Protocol & AI Readiness

- **HTNG compliance:** Full hospitality technology messaging protocol support for PMS integration
- **FIAS readiness:** Property interface adapter support for legacy hotel systems
- **Cielo Epic AI:** Edge ML model execution for occupancy prediction, anomaly detection, and predictive HVAC optimization — all on-device, zero cloud latency
- **Antenna placement:** RF zones enforced; minimum 10mm keep-out from ground plane edges

Actuation Circuit Design

Actuation reliability is a safety-critical requirement — lock failures in hospitality and healthcare environments carry significant liability. Every actuation circuit is designed with deterministic behavior under fault conditions.

1

Lock Motor Drive

H-bridge motor driver with current limiting and stall detection. Soft-start ramp reduces inrush current. Fail-secure and fail-safe modes configurable via firmware register.

2

Relay Control

Optoisolated relay drivers for HVAC and auxiliary loads. Mechanical and solid-state relay support. Status feedback via isolated GPIO for closed-loop confirmation.

3

Flyback Suppression

Schottky flyback diodes on all inductive loads. TVS arrays on relay coil lines. Switching transients validated to remain below 1V overshoot at motor terminals.

4

LC Filtering

Pi-filter networks on motor supply rails suppress switching noise re-entering the analog domain. Filter cutoff frequency tuned to motor PWM frequency harmonic rejection.

EMI Management & Compliance Certification

Achieving regulatory certification on first submission requires proactive EMI design — not reactive rework. The Caleido Xenia device is engineered to meet hospitality-grade and healthcare-adjacent compliance targets simultaneously.

Shielding Strategy

- Full-perimeter shielding cans on RF modules (WiFi, BLE, Zigbee)
- Board-level partial shields over switching power converters
- Conductive gaskets on all seams with metal enclosure variants
- Shield grounding via dedicated pad arrays — not signal ground
- Pre-scan simulation in Ansys HFSS before first prototype build

Certification Targets

- **FCC Part 15 Class B** — commercial deployment in all US hospitality venues
- **CE / RED** — European Union wireless device directive
- **UL 294** — access control systems standard for lock integration
- **IEC 62368-1** — audio/video and IT equipment safety (hospital-adjacent)
- **RoHS / REACH** — ESG-consistent materials compliance

Compliance Target: First-pass certification on FCC and CE. Pre-scan testing mandated at Rev B prototype stage. Rework after tape-out adds 8–12 weeks to schedule.

RAG Engineering Status Dashboard

A real-time risk-tiered view of all open engineering workstreams. Status classifications follow a strict RAG (Red-Amber-Green) protocol — reviewed weekly by the PCB engineering lead and escalated to the board when any Red item is unresolved beyond two sprint cycles.

Workstream	Issue / Risk	Mitigation Action	Status
ESG Sensor Accuracy	Switching noise coupling into PM2.5 ADC front-end	Add ferrite bead + guard ring; re-route analog traces	Blocker
Ground Plane Integrity	Plane split detected under MCU-to-sensor bridge route	Re-route offending trace; DRC rule update enforced	Blocker
Antenna Placement	BLE antenna keep-out violated by ground pour encroachment	Ground pour keep-out zone extended; re-pour and verify	Risk
High-Speed Routing	USB and SPI traces exceed 10cm without length matching	Apply meander tuning; verify skew in SI simulation	Risk
Power Partitioning	Supercap bank layout and sizing complete	Validated in SPICE; thermal verified at 60°C ambient	Ready
Actuation Protection	Flyback diodes and LC filters placed and verified	Bench-tested at full load; waveforms within spec	Ready
FCC Pre-Scan Readiness	Pre-scan scheduled for Rev B; RF shielding cans sourced	Shield can footprints placed; gasket spec confirmed	Ready

Engineering Governance Model

Disciplined governance is the mechanism that converts engineering rigor into predictable delivery. The Caleido Xenia program operates a three-tier governance structure — weekly execution reviews, vendor compliance gates, and manufacturing readiness checkpoints.

Weekly PCB Design Review

Structured review of all open DRC violations, RAG status updates, and schematic changes. Action items logged in Jira with owner and due date. Escalation threshold: any Red item unresolved beyond 2 cycles.

Vendor Compliance Checklist

All component vendors must submit: RoHS/REACH certificates, AEC-Q qualification data, and long-term availability commitments. Single-source components flagged for dual-source qualification before MP release.

DFM / DFT Readiness Gates

Design for Manufacturability review conducted at Rev A tape-out. Design for Testability — including boundary scan, test point coverage ≥85%, and ICT net accessibility — verified before Rev B release to CM.

Strategic Value: Hardware as a Platform Multiplier

BOARD RECOMMENDATION

The Caleido Xenia PCB is not a commodity hardware component — it is the physical foundation of a defensible ESG and AI platform. Precision hardware engineering directly amplifies platform valuation, regulatory credibility, and enterprise contract defensibility.

ESG Platform Accuracy

Certified, traceable sensor data enables audit-grade ESG reporting. Hardware precision is the competitive moat competitors cannot easily replicate in software alone.

AI-Ready Infrastructure

On-device Cielo Epic inference capability positions Caleido Xenia as an edge AI platform — not a connected sensor. This unlocks premium SaaS pricing tiers and AI feature velocity.

Hardware → Platform Valuation

Recurring ESG data subscriptions, AI analytics licensing, and compliance-as-a-service revenue streams are only possible on top of verified, reliable hardware. The PCB playbook is the foundation of the platform multiple.